

CLASS 7 MATHEMATICS

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Formula Sheets
Derivations · Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 1: Integers

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Understanding of integers and the number line from Class 6.
- Comfort with addition and subtraction of positive and negative numbers.

1.1 Properties of Addition & Subtraction of Integers

🕒 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Integers are closed, commutative, and associative under addition, and zero acts as the additive identity — these properties make integer arithmetic predictable and flexible.

Real-Life Example

Rearranging a list of daily temperature changes (some positive, some negative) before adding them up to find the net change uses the commutative and associative properties of integers.

Detailed Explanation

Definition

A property of integer addition describes a fixed rule the operation always follows, such as closure (sum of two integers is always an integer), commutativity (order doesn't matter), or associativity (grouping doesn't matter).

Key Idea

Integers are closed under both addition AND subtraction (unlike whole numbers, which are not closed under subtraction) — this is one of the key reasons integers were introduced as an extension of whole numbers.

Working Principle

Using commutativity and associativity, a long string of integer additions can be reordered and regrouped to combine positive and negative numbers separately first, making mental calculation much easier.

Applications

- Simplifying long addition/subtraction chains of positive and negative numbers in banking, temperature, or elevation problems.
- Foundation for algebraic manipulation of expressions involving signed numbers.

✓ Points to Remember (Exam Focus)

- ✓ Closure: $a + b$ and $a - b$ are always integers, for any integers a and b .
- ✓ Commutative (addition only): $a + b = b + a$; NOT true for subtraction ($a - b \neq b - a$ in general).

- ✓ Associative (addition only): $(a+b)+c = a+(b+c)$; NOT true for subtraction.
- ✓ Additive identity: $a + 0 = a$, for any integer a .

⚠ Common Student Mistakes

- ⚠ Assuming subtraction of integers is commutative or associative like addition — it is neither.
- ⚠ Forgetting integers are closed under subtraction, unlike whole numbers.

🧠 Memory Hook

'Addition plays fair (commutative, associative); Subtraction does not.'

🚀 Advanced Insights

The fact that integers are closed under both addition and subtraction (forming what mathematicians call a 'group' under addition) is a foundational idea that is formally studied in abstract algebra in much higher classes.

1.2 Multiplication & Division of Integers

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🗣 Quick Summary

Multiplying or dividing two integers with the same sign gives a positive result, while different signs give a negative result — a simple rule that governs all integer multiplication and division.

🌍 Real-Life Example

If a shop loses ₹50 per day for 6 days, the total loss is $6 \times (-50) = -300$, i.e., a loss of ₹300 — a direct real-world use of multiplying integers with different signs.

📖 Detailed Explanation

Definition

Multiplication of integers combines repeated addition (for positive multipliers) or repeated subtraction (for negative multipliers) of a signed number; division of integers is the inverse operation of multiplication, following the same sign rules.

Key Idea

Same signs ($++$ or $--$) give a positive product; different signs ($+-$ or $-+$) give a negative product — this single rule applies identically to both multiplication and division.

Working Principle

To multiply or divide two integers, first multiply/divide their absolute values, then apply the sign rule (same signs $\rightarrow$ positive result; different signs $\rightarrow$ negative result) to the final answer.

Applications

- Calculating total loss/gain over multiple days or transactions using multiplication of negative numbers.
- Solving real-world problems involving rates of change, like temperature drops over several hours.

✓ Points to Remember (Exam Focus)

- ✓ $(+) \times (+) = (+)$; $(-) \times (-) = (+)$; $(+) \times (-) = (-)$; $(-) \times (+) = (-)$.
- ✓ Same rule applies for division: same signs give positive quotient, different signs give negative quotient.
- ✓ Integers are NOT closed under division (e.g., $3 \div 2$ is not an integer).
- ✓ Division by zero is undefined for any integer.

⚠ Common Student Mistakes

- ⚠ Forgetting that two negatives multiplied together give a positive result, not a negative one.
- ⚠ Assuming integers are closed under division just because they are closed under multiplication.

**Memory Hook**

'Same signs, Positive prize; Different signs, Negative surprise.'

**Advanced Insights**

The rule ' $(-) \times (-) = (+)$ ' can actually be proven rigorously using the distributive property applied to $(-1) \times (-1)$, rather than just being an arbitrary convention — a nice bridge between arithmetic rules and formal algebraic proof.

Formula Section** Sign Rule for Integer Multiplication/Division**

Difficulty	● Easy ★☆☆
Formula	$(+) \times (+) = +$, $(-) \times (-) = +$, $(+) \times (-) = -$, $(-) \times (+) = -$
Meaning of Variables	Signs of the two integers being multiplied or divided.
When to Use	Whenever multiplying or dividing two signed integers, to determine the sign of the result before computing its value.

Common Mistakes	Mixing up the same-sign and different-sign rules, especially with three or more numbers multiplied together.
Shortcut / Memory Trick	Count the number of negative signs: an EVEN count gives a positive result; an ODD count gives a negative result.

II Quick Recap — Topics Covered So Far

- Properties of Addition & Subtraction of Integers — Integers are closed, commutative, and associative under addition, and zero acts as the additive identity — these properties make integer arithmetic predictable and flexible.
- Multiplication & Division of Integers — Multiplying or dividing two integers with the same sign gives a positive result, while different signs give a negative result — a simple rule that governs all integer multiplication and division.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Properties of Addition & Subtraction of Integers: 'Addition plays fair (commutative, associative); Subtraction does not.'
- Multiplication & Division of Integers: 'Same signs, Positive prize; Different signs, Negative surprise.'

All Common Mistakes

-  Assuming subtraction of integers is commutative or associative like addition — it is neither.
-  Forgetting integers are closed under subtraction, unlike whole numbers.
-  Forgetting that two negatives multiplied together give a positive result, not a negative one.
-  Assuming integers are closed under division just because they are closed under multiplication.

Formula Sheet

Formula	Meaning / Use
Sign Rule for Integer Multiplication/Division: $(+)\times(+) = +$, $(-)\times(-) = +$, $(+)\times(-) = -$, $(-)\times(+) = -$	Whenever multiplying or dividing two signed integers, to determine the sign of the result before computing its value.

Quick Revision Section

- Properties of Addition & Subtraction of Integers: Integers are closed, commutative, and associative under addition, and zero acts as the additive identity — these properties make integer arithmetic predictable and flexible.
- Multiplication & Division of Integers: Multiplying or dividing two integers with the same sign gives a positive result, while different signs give a negative result — a simple rule that governs all integer multiplication and division.

Chapter 2: Fractions and Decimals

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Addition and subtraction of fractions and decimals from Class 6.
- Understanding of place value in decimal numbers.

2.1 Multiplication & Division of Fractions

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Fractions are multiplied by multiplying numerators together and denominators together; dividing by a fraction is the same as multiplying by its reciprocal.

Real-Life Example

Finding $\frac{2}{3}$ of a $\frac{3}{4}$ litre bottle of juice means multiplying $\frac{2}{3} \times \frac{3}{4}$, directly giving the amount of juice in litres.

Detailed Explanation

Definition

Multiplication of fractions combines the numerators and denominators of two (or more) fractions directly; division of a fraction by another fraction is performed by multiplying the first fraction by the reciprocal (inverse) of the second.

Key Idea

'Of' in a fraction word problem (like ' $\frac{2}{3}$ of a quantity') always signals multiplication — this is a key language cue for solving fraction word problems correctly.

Working Principle

To multiply fractions: multiply numerators together and denominators together, then simplify. To divide by a fraction: flip the second fraction (find its reciprocal) and multiply instead.

Applications

- Scaling recipes up or down by a fractional amount.
- Dividing a quantity of material or land into fractional shares.

✓ Points to Remember (Exam Focus)

- ✓ $\frac{a}{b} \times \frac{c}{d} = \frac{(a \times c)}{(b \times d)}$.
- ✓ $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ (multiply by the reciprocal of the divisor).

- ✓ The reciprocal of a fraction a/b ($a \neq 0$) is b/a ; multiplying a fraction by its reciprocal always gives 1.
- ✓ 'Of' in word problems involving fractions means multiply.

⚠ Common Student Mistakes

- ⚠ Trying to find a common denominator before multiplying fractions (unlike addition, this is unnecessary for multiplication).
- ⚠ Forgetting to flip (take reciprocal of) the **SECOND** fraction only when dividing, not the first.

🧠 Memory Hook

'Multiply straight across; to Divide, flip and multiply.'

🚀 Advanced Insights

The 'flip and multiply' rule for division works because dividing by a number is mathematically defined as multiplying by its multiplicative inverse (reciprocal) — the same deep principle that underlies division in all number systems, including algebraic fractions studied later.

Formula Section

📐 Multiplication of Fractions

Difficulty	● Easy ★☆☆
Formula	$a/b \times c/d = (a \times c) / (b \times d)$
Meaning of Variables	a, c = numerators; b, d = denominators of the two fractions.
When to Use	Whenever multiplying two or more fractions together.
Common Mistakes	Accidentally cross-multiplying (as done when checking proportions) instead of multiplying straight across.
Shortcut / Memory Trick	Cancel common factors between any numerator and any denominator BEFORE multiplying, to keep numbers small.

2.2 Multiplication & Division of Decimals

🕒 Estimated time: 15 mins Difficulty: ● Medium ★☆☆ Exam Importance: ★★★★★

🔍 Quick Summary

Decimals are multiplied like whole numbers first, then the decimal point is placed based on the total number of decimal places in both numbers; dividing by a decimal is simplified by converting the divisor into a whole number first.

Real-Life Example

Calculating the cost of 2.5 kg of rice at ₹42.50 per kg requires multiplying two decimal numbers together, then correctly placing the decimal point in the answer.

Detailed Explanation

Definition

Multiplication of decimals involves multiplying the numbers as if they were whole numbers, then inserting the decimal point so the result has the same total number of decimal places as both factors combined; division of decimals often involves converting the divisor to a whole number by shifting decimal points in both the dividend and divisor equally.

Key Idea

When multiplying decimals, ignore the decimal points and multiply as whole numbers first — only at the very end do you count total decimal places (from both numbers combined) and place the decimal point that many places from the right.

Working Principle

For division by a decimal, multiply both the dividend and divisor by the same power of 10 (shifting decimal points equally in both) until the divisor becomes a whole number, then perform ordinary long division.

Applications

- Calculating costs, areas, and other real-world quantities involving decimal prices or measurements.
- Unit conversions that often involve multiplying or dividing by decimal conversion factors.

✓ Points to Remember (Exam Focus)

- ✓ To multiply decimals: multiply as whole numbers, then count total decimal places in both factors combined to place the decimal point in the product.
- ✓ To divide by a decimal: shift the decimal point in BOTH divisor and dividend equally (multiply both by the same power of 10) until the divisor is a whole number.
- ✓ Multiplying a decimal by 10, 100, or 1000 simply shifts the decimal point right by 1, 2, or 3 places respectively.
- ✓ Dividing a decimal by 10, 100, or 1000 shifts the decimal point left by 1, 2, or 3 places respectively.

Common Student Mistakes

-  Placing the decimal point in the product based on just one factor's decimal places, forgetting to combine both.

 Shifting the decimal point in only the divisor (not the dividend) when preparing for division, which changes the actual value of the answer.

Memory Hook

'Multiply first, decimal point LAST — count total places from both numbers.'

Advanced Insights

This method of shifting decimal points equally in both dividend and divisor works because multiplying both the numerator and denominator of a division by the same number never changes the actual value of the quotient — the same principle behind equivalent fractions.

II Quick Recap — Topics Covered So Far

- Multiplication & Division of Fractions — Fractions are multiplied by multiplying numerators together and denominators together; dividing by a fraction is the same as multiplying by its reciprocal.
- Multiplication & Division of Decimals — Decimals are multiplied like whole numbers first, then the decimal point is placed based on the total number of decimal places in both numbers; dividing by a decimal is simplified by converting the divisor into a whole number first.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Multiplication & Division of Fractions: 'Multiply straight across; to Divide, flip and multiply.'
- Multiplication & Division of Decimals: 'Multiply first, decimal point LAST — count total places from both numbers.'

All Common Mistakes

-  Trying to find a common denominator before multiplying fractions (unlike addition, this is unnecessary for multiplication).
-  Forgetting to flip (take reciprocal of) the SECOND fraction only when dividing, not the first.
-  Placing the decimal point in the product based on just one factor's decimal places, forgetting to combine both.
-  Shifting the decimal point in only the divisor (not the dividend) when preparing for division, which changes the actual value of the answer.

Formula Sheet

Formula	Meaning / Use
Multiplication of Fractions: $a/b \times c/d = (a \times c) / (b \times d)$	Whenever multiplying two or more fractions together.

Quick Revision Section

- Multiplication & Division of Fractions: Fractions are multiplied by multiplying numerators together and denominators together; dividing by a fraction is the same as multiplying by its reciprocal.
- Multiplication & Division of Decimals: Decimals are multiplied like whole numbers first, then the decimal point is placed based on the total number of decimal places in both numbers; dividing by a decimal is simplified by converting the divisor into a whole number first.

Chapter 3: Data Handling

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic idea of organising data using tally marks and frequency tables from Class 6.
- Comfort with addition and division of whole numbers.

3.1 Mean, Median & Mode

Estimated time: 18 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

Mean, median, and mode are three different measures of central tendency, each representing a 'typical' or central value of a data set in a different way.

Real-Life Example

A teacher calculating the average (mean) marks of a class, or finding the middle (median) score when ranking students, are everyday uses of these measures.

Detailed Explanation

Definition

Mean (average) is the sum of all observations divided by the number of observations; median is the middle value of a data set when arranged in ascending or descending order; mode is the value that occurs most frequently in a data set.

Key Idea

Mean, median, and mode can all be different values for the same data set, and each is useful in different situations — mean is sensitive to extreme values, while median and mode are not.

Working Principle

To find the median, first arrange all data in order, then pick the middle value (or average the two middle values if there is an even number of observations); to find the mode, count the frequency of each value and pick the most frequent one.

Applications

- Reporting average performance (mean marks, mean rainfall, mean temperature).
- Finding the most common shoe size sold (mode) or the middle income in a survey (median), which can be more representative than the mean in some cases.

Points to Remember (Exam Focus)

- ✓ Mean = (sum of all observations) / (number of observations).
- ✓ Median: middle value after arranging data in order; for even number of values, median = average of the two middle values.
- ✓ Mode: the most frequently occurring value in the data set (a data set can have more than one mode, or none at all).
- ✓ Range = highest value – lowest value, a simple measure of spread often studied alongside these three.

⚠ Common Student Mistakes

- ⚠ Forgetting to arrange data in order before finding the median.
- ⚠ Confusing mode (most frequent value) with mean (average value) — they can be very different numbers.



Memory Hook

'MEAN = add & divide; MEDIAN = middle after ordering; MODE = most often.'



Advanced Insights

The mean can be heavily distorted by a single extreme value (an outlier) — for example, average income in a small group can look artificially high if one person earns far more than everyone else, which is exactly why median income is often reported instead in economic statistics.

Formula Section



Mean (Arithmetic Average)

Difficulty	● Easy ★☆☆
Formula	Mean = (Sum of all observations) / (Number of observations)
Meaning of Variables	Sum of observations = total of all data values; number of observations = count of data points.
When to Use	Whenever a single 'typical' representative value for an entire data set is needed, and extreme values are not a major concern.
Common Mistakes	Dividing by the wrong count (e.g., using the sum of frequencies instead of the number of distinct observations, or vice versa).
Shortcut / Memory Trick	Double-check by ensuring the mean always lies between the lowest and highest values in the data set.

3.2 Probability — A Basic Idea

 **Estimated time:** 12 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★☆☆

Quick Summary

Probability measures how likely an event is to happen, expressed as a fraction, decimal, or percentage between 0 (impossible) and 1 (certain).

Real-Life Example

The chance of getting 'heads' when tossing a fair coin is $\frac{1}{2}$, since there are two equally likely outcomes and only one of them is heads.

Detailed Explanation

Definition

Probability of an event is defined as the number of favourable outcomes divided by the total number of equally likely possible outcomes, giving a value between 0 and 1.

Key Idea

A probability of 0 means the event is impossible, a probability of 1 means the event is certain, and everything in between represents varying degrees of likelihood.

Working Principle

To find the probability of a simple event (like getting a specific number on a die), count the number of favourable outcomes and divide by the total number of equally likely outcomes possible.

Applications

- Estimating chances in games involving dice, coins, or cards.
- Basis for understanding risk and likelihood in weather forecasting (like '70% chance of rain') and other real-world predictions.

✓ Points to Remember (Exam Focus)

- ✓ Probability of event = $\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$.
- ✓ Probability always lies between 0 and 1 (inclusive); it can also be written as a percentage (0% to 100%).
- ✓ Probability of a die showing any specific number (1 to 6) = $\frac{1}{6}$, since there are 6 equally likely outcomes.
- ✓ Sum of probabilities of all possible outcomes of an experiment always equals 1.

Common Student Mistakes

-  Calculating probability using outcomes that are not equally likely, without adjusting the method.
-  Expressing probability as a value greater than 1 or less than 0, which is never possible.

Memory Hook

'Probability = Favourable ÷ Total — always between 0 (never) and 1 (always).'

Advanced Insights

This 'classical' definition of probability (favourable outcomes ÷ total outcomes) works well for simple, equally likely situations like dice and coins, but for many real-world situations (like weather prediction), probability is instead estimated using large amounts of past data — a different, more advanced approach called experimental or statistical probability.

Formula Section

Basic Probability

Difficulty	● Easy ★☆☆
Formula	$P(\text{Event}) = (\text{Favourable outcomes}) / (\text{Total possible outcomes})$
Meaning of Variables	$P(\text{Event})$ = probability of the event happening; favourable outcomes = outcomes where the event occurs; total possible outcomes = all equally likely outcomes.
When to Use	To calculate the likelihood of a simple event in situations with equally likely outcomes, like dice, coins, or cards.
Common Mistakes	Using outcomes that aren't actually equally likely without correcting the calculation method.
Shortcut / Memory Trick	List out ALL possible outcomes first, then count how many match what you want — this avoids missing any cases.

II Quick Recap — Topics Covered So Far

- Mean, Median & Mode — Mean, median, and mode are three different measures of central tendency, each representing a 'typical' or central value of a data set in a different way.
- Probability — A Basic Idea — Probability measures how likely an event is to happen, expressed as a fraction, decimal, or percentage between 0 (impossible) and 1 (certain).

Chapter 3 — End of Chapter Revision

All Memory Hooks

- Mean, Median & Mode: 'MEAN = add & divide; MEDIAN = middle after ordering; MODE = most often.'
- Probability — A Basic Idea: 'Probability = Favourable ÷ Total — always between 0 (never) and 1 (always).'

All Common Mistakes

-  Forgetting to arrange data in order before finding the median.
-  Confusing mode (most frequent value) with mean (average value) — they can be very different numbers.
-  Calculating probability using outcomes that are not equally likely, without adjusting the method.
-  Expressing probability as a value greater than 1 or less than 0, which is never possible.

Formula Sheet

Formula	Meaning / Use
Mean (Arithmetic Average): Mean = (Sum of all observations) / (Number of observations)	Whenever a single 'typical' representative value for an entire data set is needed, and extreme values are not a major concern.
Basic Probability: $P(\text{Event}) = (\text{Favourable outcomes}) / (\text{Total possible outcomes})$	To calculate the likelihood of a simple event in situations with equally likely outcomes, like dice, coins, or cards.

Quick Revision Section

- Mean, Median & Mode: Mean, median, and mode are three different measures of central tendency, each representing a 'typical' or central value of a data set in a different way.
- Probability — A Basic Idea: Probability measures how likely an event is to happen, expressed as a fraction, decimal, or percentage between 0 (impossible) and 1 (certain).

Chapter 4: Simple Equations

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic idea of variables and expressions from Class 6 Algebra.
- Comfort with integer arithmetic.

4.1 Forming & Solving Simple Equations

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

An equation is formed by translating a word statement into a mathematical relationship using a variable, and is solved by isolating the variable using inverse operations.

Real-Life Example

If a number increased by 7 gives 15, this is written as $x + 7 = 15$, solved by subtracting 7 from both sides to get $x = 8$.

Detailed Explanation

Definition

A simple (linear) equation is a statement of equality between two expressions containing exactly one variable raised to the power 1; solving means finding the value of the variable that makes the equation true.

Key Idea

To solve an equation, perform the same operation on both sides to gradually isolate the variable — this keeps the equation balanced, like a see-saw.

Working Principle

Working step-by-step, use inverse operations (subtraction to undo addition, division to undo multiplication) applied equally to both sides, until only the variable remains on one side.

Applications

- Solving word problems involving unknown ages, quantities, or costs.
- Foundation for solving more advanced equations in higher classes (quadratic, simultaneous equations).

✓ Points to Remember (Exam Focus)

- ✓ An equation always has an equals sign; whatever operation is done to one side must be done to the other.
- ✓ To solve $ax + b = c$: first subtract b from both sides, then divide both sides by a .

- ✓ Always verify the solution by substituting it back into the original equation.
- ✓ The same value can be transposed across the equals sign by reversing its sign/operation (a shortcut for the 'do the same to both sides' rule).

⚠ Common Student Mistakes

- ⚠ Forgetting to change the sign of a term when transposing it across the equals sign.
- ⚠ Performing an operation on only one side of the equation, breaking the balance.

🧠 Memory Hook

'Balance the see-saw — same operation on both sides.'

🚀 Advanced Insights

Transposition (moving a term across the equals sign by flipping its sign or operation) is simply a shortcut for applying the same inverse operation to both sides — it looks different, but it's mathematically identical to the full balance method.

Formula Section

📐 Solving $ax + b = c$

Difficulty	● Medium ★★☆☆
Formula	$x = (c - b) / a$
Meaning of Variables	a = coefficient of x , b = constant term, c = value equation equals, x = unknown.
When to Use	For any simple linear equation with one variable, to directly compute its solution.
Common Mistakes	Dividing before subtracting, which gives the wrong order of operations and an incorrect answer.
Shortcut / Memory Trick	Undo addition/subtraction FIRST, then undo multiplication/division — reverse PEMDAS order.

4.2 Applications of Simple Equations — Word Problems

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🗨 Quick Summary

Real-world situations involving unknown quantities can be translated into simple equations, solved, and then interpreted back in the context of the original problem.



Real-Life Example

'The sum of a number and twice its value is 24' translates directly to the equation $x + 2x = 24$, solved to find $x = 8$.



Detailed Explanation

Definition

A word problem is a real-life scenario described in words that can be translated into a mathematical equation using a variable to represent the unknown quantity, then solved and interpreted.

Key Idea

The key skill is correctly translating words into algebraic symbols — phrases like 'sum of', 'twice', 'more than', and 'less than' each correspond to specific mathematical operations.

Working Principle

Read the problem carefully, assign a variable to the unknown quantity, translate the given relationship into an equation, solve the equation, and finally interpret the numeric answer back in the context of the original question.

Applications

- Solving practical problems involving ages, money, distances, and quantities in daily life and competitive exams.
- Building the problem-solving skill of translating real situations into mathematical models, used throughout higher mathematics.

✓ Points to Remember (Exam Focus)

- ✓ 'Sum of' means addition (+); 'difference' means subtraction (-); 'product' means multiplication (×); 'twice/thrice' means multiply by 2/3.
- ✓ Always define what the variable represents clearly before forming the equation.
- ✓ After solving, check whether the answer makes sense in the real-world context (e.g., age or count cannot be negative).

⚠ Common Student Mistakes

- ⚠ Misreading the word problem and translating 'less than' or 'more than' in the wrong order (e.g., '5 less than x' is $x - 5$, not $5 - x$).
- ⚠ Forgetting to check whether the final numeric answer is sensible in the given real-world context.

 Memory Hook

'Translate the words carefully — the ORDER of subtraction phrases especially matters!'

 Advanced Insights

This translation skill (words $\rightarrow$ equations) is the very foundation of 'mathematical modelling', a technique used throughout science, economics, and engineering to represent and solve real-world problems using mathematics.

II Quick Recap — Topics Covered So Far

- **Forming & Solving Simple Equations** — An equation is formed by translating a word statement into a mathematical relationship using a variable, and is solved by isolating the variable using inverse operations.
- **Applications of Simple Equations — Word Problems** — Real-world situations involving unknown quantities can be translated into simple equations, solved, and then interpreted back in the context of the original problem.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Forming & Solving Simple Equations: 'Balance the see-saw — same operation on both sides.'
- Applications of Simple Equations — Word Problems: 'Translate the words carefully — the ORDER of subtraction phrases especially matters!'

All Common Mistakes

-  Forgetting to change the sign of a term when transposing it across the equals sign.
-  Performing an operation on only one side of the equation, breaking the balance.
-  Misreading the word problem and translating 'less than' or 'more than' in the wrong order (e.g., '5 less than x' is $x - 5$, not $5 - x$).
-  Forgetting to check whether the final numeric answer is sensible in the given real-world context.

Formula Sheet

Formula	Meaning / Use
Solving $ax + b = c$: $x = (c - b) / a$	For any simple linear equation with one variable, to directly compute its solution.

Quick Revision Section

- Forming & Solving Simple Equations: An equation is formed by translating a word statement into a mathematical relationship using a variable, and is solved by isolating the variable using inverse operations.
- Applications of Simple Equations — Word Problems: Real-world situations involving unknown quantities can be translated into simple equations, solved, and then interpreted back in the context of the original problem.

Chapter 5: Lines and Angles

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic geometrical ideas — points, lines, rays, and angles from Class 6.
- Ability to measure angles using a protractor.

5.1 Related Angles — Complementary, Supplementary, Adjacent

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Pairs of angles are classified as complementary (sum to 90°), supplementary (sum to 180°), or adjacent (share a common vertex and arm), depending on their relationship.

Real-Life Example

The two acute angles in a right-angled set-square are complementary (they add up to 90°), while the angles on a straight railway line crossing a road often form supplementary pairs.

Detailed Explanation

Definition

Complementary angles are two angles whose measures add up to 90° ; supplementary angles are two angles whose measures add up to 180° ; adjacent angles share a common vertex and a common arm, with no overlap between them.

Key Idea

Complementary and supplementary relationships depend only on the SUM of the two angle measures, regardless of whether the angles are physically adjacent to each other or placed far apart.

Working Principle

Given one angle in a complementary or supplementary pair, the other can be found directly by subtracting the known angle from 90° or 180° respectively.

Applications

- Finding unknown angles in geometric figures and real-world structures like roof trusses and road intersections.
- Foundation for the angle properties of triangles and parallel lines studied later in the chapter.

✓ Points to Remember (Exam Focus)

- ✓ Complementary angles: sum = 90° (e.g., 30° and 60°).

- ✓ Supplementary angles: sum = 180° (e.g., 110° and 70°).
- ✓ Adjacent angles: share a common vertex and a common arm, and do not overlap.
- ✓ Linear pair: two adjacent angles that are also supplementary, formed when one line stands on another, together summing to 180° .

⚠ Common Student Mistakes

- ⚠ Confusing complementary (90°) with supplementary (180°) — a very common mix-up.
- ⚠ Assuming any two angles that add up to 90° or 180° must also be adjacent — they don't need to be adjacent to be complementary/supplementary.

🧠 Memory Hook

'C for Complementary, C for Corner (90°); S for Supplementary, S for Straight line (180°).'

🚀 Advanced Insights

A linear pair is always supplementary, but not every supplementary pair forms a linear pair — a linear pair specifically requires the two angles to be adjacent AND formed by one ray standing on a straight line.

5.2 Parallel Lines & a Transversal

🕒 Estimated time: 18 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔍 Quick Summary

When a transversal line crosses two parallel lines, it creates several pairs of equal or supplementary angles, following specific, predictable rules.

🌍 Real-Life Example

Railway tracks (parallel lines) crossed by a diagonal road (transversal) create multiple angle pairs that follow these exact geometric rules.

📖 Detailed Explanation

Definition

A transversal is a line that intersects two or more other lines at distinct points; when a transversal crosses two parallel lines, it creates pairs of corresponding angles, alternate interior angles, and co-interior (allied) angles with specific equal or supplementary relationships.

Key Idea

When the two lines cut by a transversal are PARALLEL, corresponding angles are equal, alternate angles are equal, and co-interior (allied) angles are supplementary — these rules only strictly hold when the lines are truly parallel.

Working Principle

By identifying the position of two angles relative to the transversal and the two lines (same side/opposite side, interior/exterior), you can determine which angle relationship rule applies and calculate unknown angles accordingly.

Applications

- Determining unknown angles in structures like bridges, staircases, and building designs that use parallel beams.
- Foundation for proving properties of triangles and quadrilaterals using parallel line properties.

✓ Points to Remember (Exam Focus)

- ✓ Corresponding angles (same position at each intersection): equal, when lines are parallel.
- ✓ Alternate interior angles (opposite sides of transversal, between the lines): equal, when lines are parallel.
- ✓ Co-interior/allied angles (same side of transversal, between the lines): supplementary (sum to 180°), when lines are parallel.
- ✓ These angle relationships can also be used in reverse: if corresponding or alternate angles are equal, the two lines MUST be parallel.

⚠ Common Student Mistakes

- ⚠ Applying these equal-angle rules to lines that are NOT actually parallel — the rules only hold for parallel lines.
- ⚠ Confusing corresponding angles (equal, same relative position) with co-interior angles (supplementary, not equal).



Memory Hook

'Corresponding = Copy (equal); Alternate = also equal, just on opposite sides; Co-interior = Cooperate to make 180° .'



Advanced Insights

The converse of these angle rules (equal corresponding/alternate angles imply the lines are parallel) is actually how surveyors and engineers verify that two structural lines are truly parallel using only angle measurements, without needing to check the entire length of both lines.

II Quick Recap — Topics Covered So Far

- Related Angles — Complementary, Supplementary, Adjacent — Pairs of angles are classified as complementary (sum to 90°), supplementary (sum to 180°), or adjacent (share a common vertex and arm), depending on their relationship.
- Parallel Lines & a Transversal — When a transversal line crosses two parallel lines, it creates several pairs of equal or supplementary angles, following specific, predictable rules.

Chapter 5 — End of Chapter Revision

All Memory Hooks

- Related Angles — Complementary, Supplementary, Adjacent: 'C for Complementary, C for Corner (90°); S for Supplementary, S for Straight line (180°).'
- Parallel Lines & a Transversal: 'Corresponding = Copy (equal); Alternate = also equal, just on opposite sides; Co-interior = Cooperate to make 180° .'

All Common Mistakes

-  Confusing complementary (90°) with supplementary (180°) — a very common mix-up.
-  Assuming any two angles that add up to 90° or 180° must also be adjacent — they don't need to be adjacent to be complementary/supplementary.
-  Applying these equal-angle rules to lines that are NOT actually parallel — the rules only hold for parallel lines.
-  Confusing corresponding angles (equal, same relative position) with co-interior angles (supplementary, not equal).

Quick Revision Section

- Related Angles — Complementary, Supplementary, Adjacent: Pairs of angles are classified as complementary (sum to 90°), supplementary (sum to 180°), or adjacent (share a common vertex and arm), depending on their relationship.
- Parallel Lines & a Transversal: When a transversal line crosses two parallel lines, it creates several pairs of equal or supplementary angles, following specific, predictable rules.

Chapter 6: The Triangle and Its Properties

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic classification of triangles from Class 6.
- Understanding of angle sum property (180° in a triangle).

6.1 Median, Altitude & the Angle Sum Property

 **Estimated time:** 15 mins **Difficulty:**  Medium **Exam Importance:** ★★☆☆     

Quick Summary

A median connects a vertex to the midpoint of the opposite side, an altitude is the perpendicular from a vertex to the opposite side, and the sum of a triangle's interior angles is always 180° .

Real-Life Example

A tent's central supporting pole, standing perpendicular from the top to the base, models an altitude of a triangular cross-section.

Detailed Explanation

Definition

A median of a triangle is a line segment joining a vertex to the midpoint of the opposite side; an altitude is a line segment from a vertex perpendicular to the line containing the opposite side; the angle sum property states that the three interior angles of any triangle always add up to 180° .

Key Idea

Every triangle has exactly three medians and three altitudes, and all three medians of a triangle always meet at a single common point (called the centroid), no matter the triangle's shape.

Working Principle

Using the angle sum property, if two angles of a triangle are known, the third can always be found by subtracting their sum from 180° ; medians and altitudes are constructed using the midpoint and perpendicularity conditions respectively.

Applications

- Structural engineering and design, where medians and altitudes help determine balance points and load distribution in triangular frameworks.
- Basis for further triangle properties like the exterior angle theorem and triangle inequality.

✓ **Points to Remember (Exam Focus)**

- ✓ Median: joins a vertex to the MIDPOINT of the opposite side.
- ✓ Altitude: perpendicular line from a vertex to the opposite side (or its extension).
- ✓ In an equilateral triangle, the median and altitude from any vertex are actually the same line segment.
- ✓ Angle sum property: sum of all three interior angles of a triangle = 180° .

⚠ Common Student Mistakes

- ⚠ Confusing a median (goes to the midpoint) with an altitude (perpendicular to the opposite side) — they are different unless the triangle is equilateral or isosceles from that specific vertex.
- ⚠ Forgetting that an altitude may fall outside the triangle for obtuse triangles, requiring the opposite side's extension.

🧠 Memory Hook

'Median = to the Middle; Altitude = at right Angle (perpendicular).'

🚀 Advanced Insights

The point where all three medians of a triangle meet (the centroid) is also the triangle's exact 'balance point' or centre of mass — if you cut a triangle out of uniform cardboard, it would balance perfectly on a pin placed at this point.

Formula Section

📐 Angle Sum Property of a Triangle

Difficulty	● Easy ★☆☆
Formula	$\angle A + \angle B + \angle C = 180^\circ$
Meaning of Variables	$\angle A, \angle B, \angle C$ = the three interior angles of the triangle.
When to Use	To find a missing angle in a triangle when the other two are known.
Common Mistakes	Adding only two angles and forgetting to include the third when checking a triangle's validity.
Shortcut / Memory Trick	Subtract the sum of the two known angles from 180° to instantly get the third.

6.2 Exterior Angle Theorem & Triangle Inequality

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

Quick Summary

The exterior angle of a triangle equals the sum of the two interior opposite (remote) angles; the triangle inequality states the sum of any two sides of a triangle must always be greater than the third side.

Real-Life Example

Checking whether three given wooden sticks (say, 3 cm, 4 cm, and 10 cm) can actually form a triangle uses the triangle inequality rule — here, $3+4=7$, which is less than 10, so they cannot form a triangle.

Detailed Explanation

Definition

An exterior angle of a triangle is formed when a side is extended beyond a vertex; the exterior angle theorem states it equals the sum of the two interior angles not adjacent to it; the triangle inequality property states that the sum of the lengths of any two sides of a triangle must be greater than the length of the third side.

Key Idea

The exterior angle theorem is a direct, quick consequence of the angle sum property, and provides a shortcut to find an exterior angle without first finding the third interior angle separately.

Working Principle

To find an exterior angle at a vertex, simply add the measures of the two interior angles that are NOT adjacent to it (the two 'remote' interior angles); to check if three lengths can form a triangle, verify that the sum of every possible pair of sides exceeds the third side.

Applications

- Quickly solving for unknown angles in triangle-based figures without extra steps.
- Checking the feasibility of triangular constructions in engineering and design (e.g., can these three given rods form a stable triangular frame?).

✓ Points to Remember (Exam Focus)

- ✓ Exterior angle = sum of the two interior opposite (remote) angles.
- ✓ An exterior angle and its adjacent interior angle are always supplementary (linear pair), summing to 180° .
- ✓ Triangle inequality: sum of any two sides $>$ third side (must hold for ALL three possible pairs of sides).
- ✓ The exterior angle is always greater than either of the two remote interior angles individually.

Common Student Mistakes

-  Adding the exterior angle's own adjacent interior angle into the exterior angle theorem calculation, instead of just the two remote (non-adjacent) angles.
-  Checking the triangle inequality for only one pair of sides instead of all three possible pairs.



Memory Hook

'Exterior angle = the two FAR (remote) interior angles added together.'



Advanced Insights

The triangle inequality isn't just a geometry rule — it's a foundational concept that reappears throughout higher mathematics (in vector spaces, complex numbers, and even abstract metric spaces) as a defining property of how 'distance' must behave.

Formula Section



Exterior Angle Theorem

Difficulty	● Medium ★★☆☆
Formula	Exterior angle = Sum of two remote interior angles
Meaning of Variables	Exterior angle = angle formed outside the triangle by extending a side; remote interior angles = the two interior angles NOT adjacent to that exterior angle.
When to Use	To quickly find an exterior angle, or a remote interior angle, without needing the third interior angle first.
Common Mistakes	Including the adjacent interior angle (linear pair partner) in the sum instead of the two remote angles.
Shortcut / Memory Trick	The exterior angle 'sees' the two angles farthest away from it inside the triangle — add those two.

II Quick Recap — Topics Covered So Far

- Median, Altitude & the Angle Sum Property — A median connects a vertex to the midpoint of the opposite side, an altitude is the perpendicular from a vertex to the opposite side, and the sum of a triangle's interior angles is always 180° .
- Exterior Angle Theorem & Triangle Inequality — The exterior angle of a triangle equals the sum of the two interior opposite (remote) angles; the triangle inequality states the sum of any two sides of a triangle must always be greater than the third side.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Median, Altitude & the Angle Sum Property: 'Median = to the Middle; Altitude = at right Angle (perpendicular).'
- Exterior Angle Theorem & Triangle Inequality: 'Exterior angle = the two FAR (remote) interior angles added together.'

All Common Mistakes

-  Confusing a median (goes to the midpoint) with an altitude (perpendicular to the opposite side) — they are different unless the triangle is equilateral or isosceles from that specific vertex.
-  Forgetting that an altitude may fall outside the triangle for obtuse triangles, requiring the opposite side's extension.
-  Adding the exterior angle's own adjacent interior angle into the exterior angle theorem calculation, instead of just the two remote (non-adjacent) angles.
-  Checking the triangle inequality for only one pair of sides instead of all three possible pairs.

Formula Sheet

Formula	Meaning / Use
Angle Sum Property of a Triangle: $\angle A + \angle B + \angle C = 180^\circ$	To find a missing angle in a triangle when the other two are known.
Exterior Angle Theorem: Exterior angle = Sum of two remote interior angles	To quickly find an exterior angle, or a remote interior angle, without needing the third interior angle first.

Quick Revision Section

- Median, Altitude & the Angle Sum Property: A median connects a vertex to the midpoint of the opposite side, an altitude is the perpendicular from a vertex to the opposite side, and the sum of a triangle's interior angles is always 180° .
- Exterior Angle Theorem & Triangle Inequality: The exterior angle of a triangle equals the sum of the two interior opposite (remote) angles; the triangle inequality states the sum of any two sides of a triangle must always be greater than the third side.

Chapter 7: Congruence of Triangles

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Understanding of triangle properties from the previous chapter.
- Ability to measure sides and angles accurately.

7.1 Congruence Criteria — SSS, SAS, ASA, RHS

 **Estimated time:** 18 mins **Difficulty:**  Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Two triangles are congruent (exactly identical in shape and size) if they satisfy one of four standard criteria: SSS, SAS, ASA, or RHS, without needing to check every single side and angle.

Real-Life Example

Identical, mass-produced parts like two same-model smartphone screens are 'congruent' — every corresponding measurement matches exactly, just like congruent triangles in geometry.

Detailed Explanation

Definition

Two triangles are congruent if all their corresponding sides and angles are exactly equal; a congruence criterion is a minimum set of matching conditions (like three sides, or two sides and the included angle) sufficient to guarantee full congruence without checking every single measurement.

Key Idea

You don't need to verify all 3 sides AND all 3 angles match to prove congruence — specific combinations of just 3 matching parts (following one of the four criteria) are always sufficient.

Working Principle

Compare the given information about two triangles against each criterion (SSS: all 3 sides match; SAS: 2 sides and the included angle match; ASA: 2 angles and the included side match; RHS: right angle, hypotenuse, and one side match for right triangles) to determine if congruence can be proven.

Applications

- Proving properties of geometric figures and structures in construction and design.
- Quality control in manufacturing, ensuring mass-produced parts are truly identical (congruent).

✓ Points to Remember (Exam Focus)

- ✓ SSS (Side-Side-Side): all three corresponding sides are equal.

- ✓ SAS (Side-Angle-Side): two sides and the INCLUDED angle (the angle between those two sides) are equal.
- ✓ ASA (Angle-Side-Angle): two angles and the INCLUDED side (the side between those two angles) are equal.
- ✓ RHS (Right angle-Hypotenuse-Side): valid only for right triangles — the hypotenuse and one other side are equal.
- ✓ SSA (two sides and a non-included angle) is NOT a valid congruence criterion in general.

⚠ Common Student Mistakes

- ⚠ Using SSA (angle not included between the two known sides) as if it were a valid criterion — it is not reliable in general.
- ⚠ Forgetting that in SAS/ASA, the angle/side must be specifically the one INCLUDED between the two given sides/angles, not just any angle/side.

🧠 Memory Hook

'SSS, SAS, ASA, RHS — the trusted four; SSA is NOT one of the core!'

🚀 Advanced Insights

AAA (all three angles equal) is explicitly NOT a valid congruence criterion — it only proves the triangles are similar (same shape, possibly different size), a distinct and important concept studied in later classes.

7.2 Applications of Congruence

🕒 **Estimated time:** 12 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔍 Quick Summary

Once two triangles are proven congruent, ALL their corresponding parts (remaining sides and angles) are automatically equal too — a powerful shortcut called CPCT.

🌍 Real-Life Example

If two triangular metal brackets are proven congruent by matching just three measurements, an engineer can be confident that every other corresponding measurement (all remaining sides and angles) also matches exactly, without measuring them individually.

📖 Detailed Explanation

Definition

CPCT (Corresponding Parts of Congruent Triangles) is the principle that once two triangles are established as congruent, every pair of their corresponding sides and angles is automatically equal.

Key Idea

CPCT is used AFTER congruence has already been proven using one of the four criteria (SSS, SAS, ASA, RHS) — it cannot be used to prove congruence itself, only to draw further conclusions from it.

Working Principle

First establish triangle congruence using a valid criterion, then use CPCT to directly conclude the equality of any other corresponding, previously-unknown side or angle in the two triangles.

Applications

- Formal geometric proofs, where CPCT is a standard, essential step to conclude equal parts after establishing congruence.
- Practical measurement problems, like finding an inaccessible distance or angle using a congruent, accessible triangle.

✓ Points to Remember (Exam Focus)

- ✓ CPCT can only be applied AFTER congruence between two triangles has already been established.
- ✓ CPCT confirms ALL remaining corresponding sides and angles are equal, not just some of them.
- ✓ Correctly identifying which vertices/sides correspond between the two triangles is essential before applying CPCT.

⚠ Common Student Mistakes

- ⚠ Using CPCT before actually proving the triangles are congruent (using it as if it were a congruence criterion itself).
- ⚠ Mismatching corresponding vertices between the two triangles, leading to incorrect CPCT conclusions.



Memory Hook

'Prove Congruence FIRST, then CPCT gives you everything else for FREE.'



Advanced Insights

CPCT is one of the most frequently used tools in formal two-column geometry proofs, and mastering exactly when and how to apply it (only after congruence, with correctly matched vertices) is essential for successfully working through more advanced geometric proofs in later classes.

II Quick Recap — Topics Covered So Far

- Congruence Criteria — SSS, SAS, ASA, RHS — Two triangles are congruent (exactly identical in shape and size) if they satisfy one of four standard criteria: SSS, SAS, ASA, or RHS, without needing to check every single side and angle.
- Applications of Congruence — Once two triangles are proven congruent, ALL their corresponding parts (remaining sides and angles) are automatically equal too — a powerful shortcut called CPCT.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Congruence Criteria — SSS, SAS, ASA, RHS: 'SSS, SAS, ASA, RHS — the trusted four; SSA is NOT one of the core!'
- Applications of Congruence: 'Prove Congruence FIRST, then CPCT gives you everything else for FREE.'

All Common Mistakes

-  Using SSA (angle not included between the two known sides) as if it were a valid criterion — it is not reliable in general.
-  Forgetting that in SAS/ASA, the angle/side must be specifically the one INCLUDED between the two given sides/angles, not just any angle/side.
-  Using CPCT before actually proving the triangles are congruent (using it as if it were a congruence criterion itself).
-  Mismatching corresponding vertices between the two triangles, leading to incorrect CPCT conclusions.

Quick Revision Section

- Congruence Criteria — SSS, SAS, ASA, RHS: Two triangles are congruent (exactly identical in shape and size) if they satisfy one of four standard criteria: SSS, SAS, ASA, or RHS, without needing to check every single side and angle.
- Applications of Congruence: Once two triangles are proven congruent, ALL their corresponding parts (remaining sides and angles) are automatically equal too — a powerful shortcut called CPCT.

Chapter 8: Comparing Quantities

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ratio and proportion concepts from Class 6.
- Comfort with percentages as a way to express fractions out of 100.

8.1 Percentages & Their Applications

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A percentage expresses a quantity as a fraction out of 100, making it easy to compare different proportions consistently across different total amounts.

Real-Life Example

A shirt marked '20% off' at a store means the discount is 20 out of every 100 rupees of the original price, making it easy to compare discounts across differently priced items.

Detailed Explanation

Definition

A percentage is a way of expressing a number as a fraction of 100, denoted using the '%' symbol; to convert a fraction to a percentage, multiply it by 100, and to convert a percentage to a fraction, divide by 100.

Key Idea

Percentages provide a common, standardised scale (always out of 100) that makes comparing different proportions or changes much easier than comparing raw fractions with different denominators.

Working Principle

To find a percentage of a quantity, convert the percentage to a fraction (divide by 100) or decimal, then multiply it by the given quantity.

Applications

- Calculating discounts, marks scored in exams, interest rates, and statistical data comparisons.
- Foundation for profit/loss and simple/compound interest calculations later in the chapter.

✓ Points to Remember (Exam Focus)

- ✓ To convert a fraction to a percentage: multiply by 100 (e.g., $\frac{3}{4} = 75\%$).
- ✓ To convert a percentage to a fraction: divide by 100 and simplify (e.g., $40\% = \frac{40}{100} = \frac{2}{5}$).
- ✓ To find $x\%$ of a quantity Q : $(\frac{x}{100}) \times Q$.

✓ Percentage increase/decrease = (change in value / original value) × 100.

⚠ Common Student Mistakes

- ⚠ Forgetting to divide by 100 when converting a percentage back into a decimal or fraction for calculation.
- ⚠ Calculating percentage change using the wrong base (using the new value instead of the ORIGINAL value in the denominator).

🧠 Memory Hook

'Percent = PER-100 — always relative to a base of 100.'

🚀 Advanced Insights

Percentage change calculations can be misleading if not interpreted carefully — for example, a 50% decrease followed by a 50% increase does NOT bring a value back to its original amount, since each percentage is calculated on a different (changing) base value.

Formula Section

📐 Percentage of a Quantity

Difficulty	● Easy ★☆☆
Formula	$x\% \text{ of } Q = (x/100) \times Q$
Meaning of Variables	x = the percentage value, Q = the total quantity.
When to Use	To find a specific percentage portion of any given quantity.
Common Mistakes	Forgetting to divide the percentage by 100 before multiplying.
Shortcut / Memory Trick	Percent literally means 'per hundred' — always divide by 100 first.

8.2 Profit, Loss & Simple Interest

🕒 Estimated time: 18 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🧠 Quick Summary

Profit or loss is calculated by comparing selling price to cost price, while simple interest is the extra money earned or paid on a principal amount over time at a fixed rate.

Real-Life Example

A shopkeeper buying a toy for ₹80 (cost price) and selling it for ₹100 (selling price) makes a profit of ₹20; a bank paying simple interest on a fixed deposit grows the deposited money predictably over time.

Detailed Explanation

Definition

Profit is the positive difference when selling price exceeds cost price; loss is the negative difference when cost price exceeds selling price; simple interest is the fixed extra amount earned or paid on a principal sum, calculated using a constant rate applied only to the original principal each year.

Key Idea

Profit/loss percentage is always calculated with respect to the COST PRICE (not the selling price), which is a frequently tested and easily confused detail.

Working Principle

Profit = Selling Price – Cost Price (if positive); Loss = Cost Price – Selling Price (if positive); Simple Interest = $(\text{Principal} \times \text{Rate} \times \text{Time}) / 100$, added to the principal to get the total amount.

Applications

- Business and trade calculations for setting prices and analysing profitability.
- Banking and finance for understanding basic loan and fixed-deposit interest calculations.

✓ Points to Remember (Exam Focus)

- ✓ Profit % = $(\text{Profit} / \text{Cost Price}) \times 100$; Loss % = $(\text{Loss} / \text{Cost Price}) \times 100$ — always based on Cost Price.
- ✓ Simple Interest (SI) = $(P \times R \times T) / 100$, where P = Principal, R = Rate of interest per annum, T = Time in years.
- ✓ Total Amount = Principal + Simple Interest.
- ✓ Selling Price = Cost Price + Profit, or Cost Price – Loss.

Common Student Mistakes

-  Calculating profit/loss percentage based on selling price instead of cost price.
-  Forgetting to divide by 100 in the simple interest formula, or mixing up rate and time values.

Memory Hook

'Profit/Loss % is always on the COST — that's the original investment.'

Advanced Insights

Unlike simple interest (calculated only on the original principal every year), compound interest — studied in later classes — calculates interest on the accumulated amount (principal + previously earned interest), causing money to grow progressively faster over time.

Formula Section

Simple Interest

Difficulty	● Medium ★★☆☆
Formula	$SI = (P \times R \times T) / 100$
Meaning of Variables	SI = simple interest, P = principal amount, R = rate of interest per annum (%), T = time in years.
When to Use	To calculate the interest earned or owed on a principal amount over a given time at a fixed annual rate.
Common Mistakes	Forgetting to divide by 100, or using time in months without converting to years.
Shortcut / Memory Trick	Remember 'PRT over 100' — multiply the three given values, then divide by 100.

II Quick Recap — Topics Covered So Far

- Percentages & Their Applications — A percentage expresses a quantity as a fraction out of 100, making it easy to compare different proportions consistently across different total amounts.
- Profit, Loss & Simple Interest — Profit or loss is calculated by comparing selling price to cost price, while simple interest is the extra money earned or paid on a principal amount over time at a fixed rate.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- Percentages & Their Applications: 'Percent = PER-100 — always relative to a base of 100.'
- Profit, Loss & Simple Interest: 'Profit/Loss % is always on the COST — that's the original investment.'

All Common Mistakes

-  Forgetting to divide by 100 when converting a percentage back into a decimal or fraction for calculation.
-  Calculating percentage change using the wrong base (using the new value instead of the ORIGINAL value in the denominator).
-  Calculating profit/loss percentage based on selling price instead of cost price.
-  Forgetting to divide by 100 in the simple interest formula, or mixing up rate and time values.

Formula Sheet

Formula	Meaning / Use
Percentage of a Quantity: $x\%$ of $Q = (x/100) \times Q$	To find a specific percentage portion of any given quantity.
Simple Interest: $SI = (P \times R \times T) / 100$	To calculate the interest earned or owed on a principal amount over a given time at a fixed annual rate.

Quick Revision Section

- Percentages & Their Applications: A percentage expresses a quantity as a fraction out of 100, making it easy to compare different proportions consistently across different total amounts.
- Profit, Loss & Simple Interest: Profit or loss is calculated by comparing selling price to cost price, while simple interest is the extra money earned or paid on a principal amount over time at a fixed rate.

Chapter 9: Rational Numbers

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Understanding of fractions and integers from earlier chapters.
- Comfort with addition, subtraction, multiplication, and division of fractions.

9.1 Introduction to Rational Numbers

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A rational number is any number that can be expressed as a fraction p/q , where p and q are integers and q is not zero — this includes all fractions, integers, and terminating/repeating decimals.

Real-Life Example

Sharing 3 chocolate bars equally among 4 friends gives each person $3/4$ of a bar — a rational number representing a precise, fair share.

Detailed Explanation

Definition

A rational number is any number that can be written in the form p/q , where p and q are integers and $q \neq 0$; every integer is also a rational number (since it can be written as itself divided by 1).

Key Idea

Rational numbers include ALL integers, all fractions, and all terminating or repeating decimals — essentially, any number that can be exactly expressed as a ratio of two whole numbers.

Working Principle

To check if a number is rational, try to express it as a fraction of two integers; if this is possible (and the denominator isn't zero), the number qualifies as rational.

Applications

- Understanding the broader number system, connecting integers, fractions, and decimals into one unified category.
- Foundation for further number system studies (like irrational numbers) in higher classes.

✓ Points to Remember (Exam Focus)

- ✓ Every integer is a rational number (e.g., $5 = 5/1$).
- ✓ A rational number's denominator can NEVER be zero (division by zero is undefined).

- ✓ Rational numbers can be positive, negative, or zero ($0 = 0/1$ is rational).
- ✓ Two rational numbers p/q and $(p \times k)/(q \times k)$ (for non-zero k) represent the same value — equivalent rational numbers.

Common Student Mistakes

-  Forgetting that ALL integers and whole numbers are also rational numbers, not a separate category.
-  Allowing a zero denominator when forming or simplifying a rational number, which is mathematically invalid.

Memory Hook

'Rational = Ratio-nal — any number that IS a ratio of two integers.'

Advanced Insights

Numbers that CANNOT be written as a simple fraction of two integers (like π or $\sqrt{2}$) are called irrational numbers — together, rational and irrational numbers make up the complete set of real numbers, a distinction explored in depth in Class 9 and beyond.

9.2 Operations on Rational Numbers

 **Estimated time:** 18 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Rational numbers are added, subtracted, multiplied, and divided using the same fraction rules learned earlier, extended to include negative numerators and denominators.

Real-Life Example

Combining a temperature change of $-3/2$ °C in the morning and $+5/2$ °C in the afternoon requires adding two rational numbers to find the net change.

Detailed Explanation

Definition

Operations on rational numbers follow the same rules as fraction operations, with careful attention to sign rules when negative rational numbers are involved.

Key Idea

To compare or add/subtract rational numbers with different denominators, first convert them to a common denominator, exactly like with regular fractions, keeping careful track of signs throughout.

Working Principle

For addition/subtraction, find the LCM of denominators and convert to like fractions before combining numerators; for multiplication, multiply numerators and denominators straight across; for division, multiply by the reciprocal — following integer sign rules throughout.

Applications

- Solving real-life problems involving fractional or negative quantities together, such as temperature changes, financial gains/losses, or measurement adjustments.
- Essential foundation for algebraic manipulation involving rational coefficients in later classes.

✓ Points to Remember (Exam Focus)

- ✓ A rational number's standard form has a positive denominator and no common factors between numerator and denominator (fully simplified).
- ✓ To find a rational number between two given rational numbers, convert both to a common denominator and pick any value in between.
- ✓ Sign rules for multiplication/division of rational numbers follow the same rules as integers (same signs → positive, different signs → negative).
- ✓ There are infinitely many rational numbers between any two given rational numbers.

⚠ Common Student Mistakes

- ⚠ Forgetting to convert a rational number to standard form (positive denominator) before comparing or ordering multiple rational numbers.
- ⚠ Applying incorrect sign rules when multiplying or dividing negative rational numbers.

**Memory Hook**

'Rational number operations = fraction rules + integer sign rules, combined.'

**Advanced Insights**

The fact that infinitely many rational numbers exist between any two given rational numbers (called 'density' of rational numbers) is a fascinating mathematical property — yet, despite this density, rational numbers still don't cover the ENTIRE number line, since irrational numbers like $\sqrt{2}$ fill in the remaining gaps.

II Quick Recap — Topics Covered So Far

- Introduction to Rational Numbers — A rational number is any number that can be expressed as a fraction p/q , where p and q are integers and q is not zero — this includes all fractions, integers, and terminating/repeating decimals.

- Operations on Rational Numbers — Rational numbers are added, subtracted, multiplied, and divided using the same fraction rules learned earlier, extended to include negative numerators and denominators.

Chapter 9 — End of Chapter Revision

All Memory Hooks

- Introduction to Rational Numbers: 'Rational = Ratio-nal — any number that IS a ratio of two integers.'
- Operations on Rational Numbers: 'Rational number operations = fraction rules + integer sign rules, combined.'

All Common Mistakes

-  Forgetting that ALL integers and whole numbers are also rational numbers, not a separate category.
-  Allowing a zero denominator when forming or simplifying a rational number, which is mathematically invalid.
-  Forgetting to convert a rational number to standard form (positive denominator) before comparing or ordering multiple rational numbers.
-  Applying incorrect sign rules when multiplying or dividing negative rational numbers.

Quick Revision Section

- Introduction to Rational Numbers: A rational number is any number that can be expressed as a fraction p/q , where p and q are integers and q is not zero — this includes all fractions, integers, and terminating/repeating decimals.
- Operations on Rational Numbers: Rational numbers are added, subtracted, multiplied, and divided using the same fraction rules learned earlier, extended to include negative numerators and denominators.

Chapter 10: Practical Geometry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic compass and ruler construction skills from Class 6.
- Understanding of triangle properties and congruence criteria.

10.1 Constructing Triangles — Given Different Conditions

 **Estimated time:** 18 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

A unique triangle can be constructed given specific combinations of measurements — three sides (SSS), two sides and an included angle (SAS), two angles and an included side (ASA), or the hypotenuse and a side of a right triangle (RHS).

Real-Life Example

An architect constructing a precise triangular roof truss uses exactly this kind of construction, building the triangle from a specific known set of side lengths and angles.

Detailed Explanation

Definition

Triangle construction is the precise drawing of a triangle using a ruler, compass, and protractor, based on a minimum sufficient set of given measurements (mirroring the congruence criteria: SSS, SAS, ASA, or RHS).

Key Idea

The specific construction method used directly corresponds to which congruence criterion applies to the given information — the same minimum information (3 sides, or 2 sides + included angle, etc.) that proves congruence is also exactly enough to construct a unique triangle.

Working Principle

Based on the type of information given (three sides, two sides and an angle, two angles and a side, or a right triangle's hypotenuse and one side), a specific step-by-step compass-and-ruler procedure is followed to accurately construct the triangle.

Applications

- Technical and architectural drawing, where triangles must be constructed to precise specifications.
- Basis for constructing more complex geometric figures built from combinations of triangles.

Points to Remember (Exam Focus)

- ✓ SSS construction: draw one side first, then use compass arcs from both endpoints (set to the other two side lengths) to locate the third vertex.
- ✓ SAS construction: draw one side, construct the included angle at one endpoint using a protractor, then mark the second given side length along the new arm.
- ✓ ASA construction: draw the given side, then construct both given angles at its two endpoints; the point where the two new arms intersect is the third vertex.
- ✓ RHS construction: draw the given side (leg), construct a right angle at one end, then use the hypotenuse length (compass arc) to locate the third vertex.

⚠ Common Student Mistakes

- ⚠ Attempting to construct a triangle using an SSA-type set of measurements, which doesn't guarantee a unique triangle.
- ⚠ Constructing the given angle at the wrong vertex/endpoint in SAS or ASA constructions.



Memory Hook

'Construction method = matches the congruence criterion given.'



Advanced Insights

Not every combination of three sides can actually form a triangle — before attempting an SSS construction, it's essential to first verify the triangle inequality (sum of any two sides $>$ third side) holds for the given lengths, otherwise the construction will fail.

II Quick Recap — Topics Covered So Far

- Constructing Triangles — Given Different Conditions — A unique triangle can be constructed given specific combinations of measurements — three sides (SSS), two sides and an included angle (SAS), two angles and an included side (ASA), or the hypotenuse and a side of a right triangle (RHS).

Chapter 10 — End of Chapter Revision

All Memory Hooks

- Constructing Triangles — Given Different Conditions: 'Construction method = matches the congruence criterion given.'

All Common Mistakes

-  Attempting to construct a triangle using an SSA-type set of measurements, which doesn't guarantee a unique triangle.
-  Constructing the given angle at the wrong vertex/endpoint in SAS or ASA constructions.

Quick Revision Section

- Constructing Triangles — Given Different Conditions: A unique triangle can be constructed given specific combinations of measurements — three sides (SSS), two sides and an included angle (SAS), two angles and an included side (ASA), or the hypotenuse and a side of a right triangle (RHS).

Chapter 11: Perimeter and Area

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Perimeter and area formulas for rectangles and squares from Class 6.
- Basic understanding of circles (radius, diameter) from earlier classes.

11.1 Area of Parallelogram & Triangle

 **Estimated time:** 15 mins **Difficulty:**  Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

The area of a parallelogram is base times height (perpendicular height, not the slant side), and the area of a triangle is exactly half of a corresponding parallelogram's area.

Real-Life Example

Calculating the area of a slanted plot of land (parallelogram-shaped) for construction requires using the perpendicular height, not the slanted side length, to get an accurate area.

Detailed Explanation

Definition

The area of a parallelogram is the product of its base and the corresponding perpendicular height (altitude); the area of a triangle is half the product of its base and corresponding perpendicular height.

Key Idea

A triangle's area formula ($\frac{1}{2} \times \text{base} \times \text{height}$) comes directly from the fact that any triangle is exactly half of a parallelogram sharing the same base and height — this connection is key to understanding, not just memorising, the formula.

Working Principle

The height used in these formulas must always be the PERPENDICULAR distance between the base and the opposite side/vertex, never the slanted or slant-side length.

Applications

- Calculating land area, floor plans, and material requirements for triangular or parallelogram-shaped surfaces.
- Foundation for area formulas of more complex polygons built from parallelograms and triangles.

✓ Points to Remember (Exam Focus)

- ✓ Area of parallelogram = base $\times$ height (perpendicular height, not the slant side).

- ✓ Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$ (perpendicular height from the base to the opposite vertex).
- ✓ Any side of a triangle or parallelogram can be chosen as the 'base', as long as the corresponding perpendicular height is used.
- ✓ A triangle's area is always exactly half of a parallelogram sharing the same base and height.

⚠ Common Student Mistakes

- ⚠ Using the slant (non-perpendicular) side length as the 'height' in area calculations instead of the true perpendicular height.
- ⚠ Forgetting the $\frac{1}{2}$ factor when calculating a triangle's area, mistakenly using the parallelogram formula instead.

🧠 Memory Hook

'Height must be PERPENDICULAR — always straight down, never slanted.'

🚀 Advanced Insights

This 'triangle = half a parallelogram' relationship is a specific example of a more general principle in geometry — many area formulas for complex shapes can be derived by relating them back to simpler shapes (rectangles, parallelograms) through decomposition or doubling.

Formula Section

📐 Area of a Triangle

Difficulty	● Medium ★★☆☆
Formula	$A = \frac{1}{2} \times \text{base} \times \text{height}$
Meaning of Variables	A = area, base = any chosen side of the triangle, height = perpendicular distance from that base to the opposite vertex.
When to Use	To find the area enclosed within any triangle, given a base and its corresponding perpendicular height.
Common Mistakes	Using a slanted side length as height instead of the true perpendicular distance.
Shortcut / Memory Trick	Picture folding the triangle to form half of a rectangle/parallelogram — that's where the $\frac{1}{2}$ comes from.

11.2 Circumference & Area of a Circle

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔍 Quick Summary

The circumference (perimeter) of a circle is found using $2\pi r$, and its area is found using πr^2 , where π is a special constant approximately equal to $22/7$ or 3.14 .

🌍 Real-Life Example

Calculating the length of fencing needed around a circular garden uses the circumference formula, while calculating how much grass seed is needed to cover it uses the area formula.

📖 Detailed Explanation

Definition

The circumference of a circle is the total distance around its boundary; the area of a circle is the total surface enclosed within its boundary; π (pi) is a mathematical constant representing the fixed ratio of a circle's circumference to its diameter, approximately $3.14159\dots$

Key Idea

The ratio of ANY circle's circumference to its diameter is always exactly the same constant value, π — this remarkable, universal consistency across circles of all sizes is what makes π such a fundamental mathematical constant.

Working Principle

Given a circle's radius (or diameter), simply substitute the value into the circumference formula ($2\pi r$) or area formula (πr^2), using $\pi \approx 22/7$ (useful when radius is a multiple of 7) or $\pi \approx 3.14$.

Applications

- Calculating fencing, boundary materials (circumference) or covering materials, paint, tiling (area) for circular objects and spaces.
- Used extensively in engineering, architecture, and design involving circular or cylindrical components.

✓ Points to Remember (Exam Focus)

- ✓ Circumference of circle = $2\pi r = \pi d$ (where d = diameter = $2r$).
- ✓ Area of circle = πr^2 .
- ✓ Use $\pi \approx 22/7$ when the radius is a multiple of 7 (for cleaner calculation); otherwise use $\pi \approx 3.14$.
- ✓ Area is always in square units (cm^2 , m^2); circumference is in linear units (cm, m), just like any perimeter.

⚠ Common Student Mistakes

- ⚠ Confusing radius and diameter — remember diameter is always twice the radius, and using the wrong one leads to answers off by a factor of 2 or 4.

⚠ Forgetting to square the radius in the area formula, or squaring it in the circumference formula by mistake.



Memory Hook

'Circumference has one 'r' ($2\pi r$); Area has 'r-squared' (πr^2) — area grows faster than circumference.'



Advanced Insights

π is an irrational number (its decimal expansion never ends or repeats), and this was actually proven mathematically centuries after it was already being used practically in construction and design — a great example of how a useful mathematical tool can be used effectively even before all its deeper properties are fully understood.

Formula Section



Circumference of a Circle

Difficulty	● Easy ★☆☆
Formula	$C = 2\pi r$
Meaning of Variables	C = circumference, $\pi \approx 22/7$ or 3.14, r = radius of the circle.
When to Use	To find the total boundary length (perimeter) of any circle.
Common Mistakes	Using diameter instead of radius without adjusting the formula ($C = \pi d$ is equivalent, but don't double-count).
Shortcut / Memory Trick	Circumference = $\pi \times$ diameter is often quicker if diameter is directly given.



Area of a Circle

Difficulty	● Easy ★☆☆
Formula	$A = \pi r^2$
Meaning of Variables	A = area, $\pi \approx 22/7$ or 3.14, r = radius of the circle.
When to Use	To find the total surface enclosed within a circle's boundary.
Common Mistakes	Forgetting to square the radius, or squaring the diameter instead of the radius.
Shortcut / Memory Trick	Always square the RADIUS specifically, never the diameter, in this formula.

II Quick Recap — Topics Covered So Far

- **Area of Parallelogram & Triangle** — The area of a parallelogram is base times height (perpendicular height, not the slant side), and the area of a triangle is exactly half of a corresponding parallelogram's area.
- **Circumference & Area of a Circle** — The circumference (perimeter) of a circle is found using $2\pi r$, and its area is found using πr^2 , where π is a special constant approximately equal to $22/7$ or 3.14 .

Chapter 11 — End of Chapter Revision

All Memory Hooks

- Area of Parallelogram & Triangle: 'Height must be PERPENDICULAR — always straight down, never slanted.'
- Circumference & Area of a Circle: 'Circumference has one 'r' ($2\pi r$); Area has 'r-squared' (πr^2) — area grows faster than circumference.'

All Common Mistakes

-  Using the slant (non-perpendicular) side length as the 'height' in area calculations instead of the true perpendicular height.
-  Forgetting the $\frac{1}{2}$ factor when calculating a triangle's area, mistakenly using the parallelogram formula instead.
-  Confusing radius and diameter — remember diameter is always twice the radius, and using the wrong one leads to answers off by a factor of 2 or 4.
-  Forgetting to square the radius in the area formula, or squaring it in the circumference formula by mistake.

Formula Sheet

Formula	Meaning / Use
Area of a Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$	To find the area enclosed within any triangle, given a base and its corresponding perpendicular height.
Circumference of a Circle: $C = 2\pi r$	To find the total boundary length (perimeter) of any circle.
Area of a Circle: $A = \pi r^2$	To find the total surface enclosed within a circle's boundary.

Quick Revision Section

- Area of Parallelogram & Triangle: The area of a parallelogram is base times height (perpendicular height, not the slant side), and the area of a triangle is exactly half of a corresponding parallelogram's area.
- Circumference & Area of a Circle: The circumference (perimeter) of a circle is found using $2\pi r$, and its area is found using πr^2 , where π is a special constant approximately equal to $22/7$ or 3.14 .

Chapter 12: Algebraic Expressions

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic idea of variables and expressions from Class 6 Algebra.
- Comfort with integer arithmetic and sign rules.

12.1 Terms, Factors, Coefficients & Like/Unlike Terms

Estimated time: 15 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

An algebraic expression is built from terms, which are combinations of factors (numbers and variables); terms with identical variable parts are called like terms, and can be combined directly.

Real-Life Example

If x represents the price of one pen, then ' $3x + 5$ ' could represent the total cost of 3 pens plus a ₹5 delivery fee — a simple real-world algebraic expression.

Detailed Explanation

Definition

A term is a single part of an algebraic expression, separated by + or – signs (e.g., in $3x + 5y - 2$, the terms are $3x$, $5y$, and -2); a coefficient is the numerical factor multiplying a variable in a term; like terms have exactly the same variable(s) raised to the same power(s), differing only in their coefficient.

Key Idea

Only LIKE terms can be combined (added or subtracted) directly by combining their coefficients — unlike terms must be kept separate in a simplified expression.

Working Principle

To simplify an algebraic expression, group all like terms together and combine their coefficients, while leaving unlike terms as they are, added or subtracted between the combined groups.

Applications

- Simplifying complex algebraic expressions before solving equations or substituting values.
- Foundation for all higher algebra, including polynomial operations in later classes.

✓ Points to Remember (Exam Focus)

- ✓ Like terms: same variable(s) with the same power(s) (e.g., $3x^2$ and $7x^2$ are like terms; $3x^2$ and $3x$ are NOT).

- ✓ Unlike terms: different variables or different powers (e.g., $3x$ and $3y$ are unlike terms).
- ✓ To combine like terms, simply add/subtract their coefficients and keep the common variable part unchanged.
- ✓ A term's coefficient can be positive, negative, or even a fraction.

⚠ Common Student Mistakes

- ⚠ Combining unlike terms directly (e.g., wrongly simplifying $3x + 2y$ as $5xy$ or $5(x+y)$).
- ⚠ Forgetting that x^2 and x are different (unlike) terms, despite sharing the same base variable.

🧠 Memory Hook

'Only LIKE terms can be combined — same variable, same power, no exceptions.'

🚀 Advanced Insights

This 'like terms' rule is exactly why algebraic expressions behave differently from simple arithmetic — you can always add two numbers together, but you can only directly combine algebraic terms when their variable structure matches exactly.

12.2 Addition & Subtraction of Algebraic Expressions

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔍 Quick Summary

Algebraic expressions are added or subtracted by combining only their corresponding like terms, keeping careful track of signs throughout, especially during subtraction.

🌍 Real-Life Example

If the total cost of pens and pencils bought on Monday is $(3x + 2y)$ and on Tuesday is $(2x + 5y)$, the combined weekly cost is found by adding these two algebraic expressions together.

📖 Detailed Explanation

Definition

Addition of algebraic expressions combines all like terms from both expressions using their sign; subtraction of algebraic expressions involves changing the sign of every term in the expression being subtracted, then combining like terms as in addition.

Key Idea

Subtracting an entire algebraic expression means changing the sign of EVERY term within it before combining — a single sign error on just one term will make the entire result incorrect.

Working Principle

Arrange the expressions with like terms aligned (either horizontally or in columns), then combine the coefficients of each group of like terms, applying subtraction's sign-flip rule carefully to every term of the expression being subtracted.

Applications

- Simplifying combined costs, quantities, or measurements expressed algebraically.
- Essential preliminary skill for solving algebraic equations and later polynomial operations.

✓ Points to Remember (Exam Focus)

- ✓ Addition: combine like terms directly, keeping their original signs.
- ✓ Subtraction: change the sign of EVERY term in the expression being subtracted, then add as usual.
- ✓ Always double-check that terms being combined are truly 'like terms' before merging their coefficients.
- ✓ The result should list any remaining unlike terms separately, without further combination.

⚠ Common Student Mistakes

- ⚠ Changing the sign of only the first term when subtracting an expression, instead of every term within it.
- ⚠ Combining unlike terms by mistake, especially when several different variables are present in a longer expression.

💡 Memory Hook

'Subtracting an expression = flipping EVERY sign inside it, then adding.'

🚀 Advanced Insights

This exact 'flip every sign, then add' technique for subtraction is a direct extension of the same distributive property rule used for subtracting a bracketed group of numbers — algebra simply applies the same core arithmetic logic to expressions with variables.

II Quick Recap — Topics Covered So Far

- Terms, Factors, Coefficients & Like/Unlike Terms — An algebraic expression is built from terms, which are combinations of factors (numbers and variables); terms with identical variable parts are called like terms, and can be combined directly.

- Addition & Subtraction of Algebraic Expressions — Algebraic expressions are added or subtracted by combining only their corresponding like terms, keeping careful track of signs throughout, especially during subtraction.

Chapter 12 — End of Chapter Revision

All Memory Hooks

- Terms, Factors, Coefficients & Like/Unlike Terms: 'Only LIKE terms can be combined — same variable, same power, no exceptions.'
- Addition & Subtraction of Algebraic Expressions: 'Subtracting an expression = flipping EVERY sign inside it, then adding.'

All Common Mistakes

-  Combining unlike terms directly (e.g., wrongly simplifying $3x + 2y$ as $5xy$ or $5(x+y)$).
-  Forgetting that x^2 and x are different (unlike) terms, despite sharing the same base variable.
-  Changing the sign of only the first term when subtracting an expression, instead of every term within it.
-  Combining unlike terms by mistake, especially when several different variables are present in a longer expression.

Quick Revision Section

- Terms, Factors, Coefficients & Like/Unlike Terms: An algebraic expression is built from terms, which are combinations of factors (numbers and variables); terms with identical variable parts are called like terms, and can be combined directly.
- Addition & Subtraction of Algebraic Expressions: Algebraic expressions are added or subtracted by combining only their corresponding like terms, keeping careful track of signs throughout, especially during subtraction.

Chapter 13: Exponents and Powers

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Comfort with repeated multiplication of numbers.
- Basic understanding of place value and large numbers.

13.1 Exponents & Laws of Exponents

 **Estimated time:** 18 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

An exponent shows how many times a base number is multiplied by itself, and a set of consistent laws governs how exponents combine during multiplication, division, and powers of powers.

Real-Life Example

Expressing the vast distance of light-years or extremely tiny sizes like bacteria in scientific notation (using powers of 10) makes such extreme numbers far easier to write and compare.

Detailed Explanation

Definition

An exponent (or power) indicates how many times a base number is multiplied by itself, written as a^n (read 'a to the power n'), where a is the base and n is the exponent.

Key Idea

The laws of exponents provide shortcuts for combining powers without expanding them fully — for example, multiplying same-base powers simply means adding their exponents ($a^m \times a^n = a^{m+n}$).

Working Principle

Each law of exponents is derived directly from the basic definition of repeated multiplication — for instance, $a^m \times a^n$ means 'a multiplied m times, then multiplied n more times', totalling $(m+n)$ multiplications of a.

Applications

- Scientific notation for extremely large or small numbers in physics, chemistry, and astronomy.
- Foundation for polynomial and algebraic manipulation with variable exponents in later classes.

✓ Points to Remember (Exam Focus)

- ✓ $a^m \times a^n = a^{m+n}$ (same base, multiplication $\rightarrow$ add exponents).
- ✓ $a^m \div a^n = a^{m-n}$ (same base, division $\rightarrow$ subtract exponents).
- ✓ $(a^m)^n = a^{mn}$ (power of a power $\rightarrow$ multiply exponents).

✓ $a^0 = 1$ (any non-zero base raised to power 0 equals 1); $a^{-n} = 1/a^n$ (negative exponent means reciprocal).

⚠ Common Student Mistakes

- ⚠ Adding exponents when the bases are actually different (the laws only apply when the base is the SAME).
- ⚠ Forgetting that a negative exponent means 'take the reciprocal', not 'make the number negative'.

🧠 Memory Hook

'Same base: Multiply → ADD exponents; Divide → SUBTRACT exponents; Power of power → MULTIPLY exponents.'

🚀 Advanced Insights

The rule $a^0 = 1$ isn't arbitrary — it follows logically from the division law: $a^n \div a^n = a^{n-n} = a^0$, but any non-zero number divided by itself also equals 1, so a^0 MUST equal 1 for the exponent laws to remain consistent.

Formula Section

📐 Laws of Exponents (Same Base)

Difficulty	● Medium ★★☆☆
Formula	$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$
Meaning of Variables	a = base (non-zero), m, n = exponents (integers).
When to Use	Whenever combining, dividing, or raising powers that share the same base.
Common Mistakes	Applying these rules when the bases are different — they only work for matching bases.
Shortcut / Memory Trick	Multiplication → Add; Division → Subtract; Power of Power → Multiply — three different operations for three different situations.

13.2 Expressing Large Numbers in Standard Form

🕒 Estimated time: 12 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔑 Quick Summary

Standard form (scientific notation) expresses very large or very small numbers compactly as a number between 1 and 10 multiplied by a power of 10.

Real-Life Example

The distance from Earth to the Sun (about 150,000,000 km) is far more conveniently written in standard form as 1.5×10^8 km.

Detailed Explanation

Definition

Standard form (scientific notation) expresses a number as $k \times 10^n$, where $1 \leq k < 10$ and n is an integer (positive for large numbers, negative for small numbers less than 1).

Key Idea

The power of 10 in standard form directly tells you how many places to shift the decimal point — a positive exponent shifts it right (making a larger number), a negative exponent shifts it left (making a smaller number).

Working Principle

To convert a large number to standard form, move the decimal point left until only one non-zero digit remains before it, and count how many places you moved — this count becomes the positive exponent of 10.

Applications

- Expressing astronomical distances, population figures, and microscopic measurements compactly in science.
- Simplifying calculations involving extremely large or small numbers in scientific and engineering contexts.

✓ Points to Remember (Exam Focus)

- ✓ Standard form: $k \times 10^n$, where $1 \leq k < 10$.
- ✓ Large numbers (≥ 10): positive exponent n ; e.g., $45,000 = 4.5 \times 10^4$.
- ✓ Small numbers (< 1): negative exponent n ; e.g., $0.00023 = 2.3 \times 10^{-4}$.
- ✓ The exponent equals the number of places the decimal point is shifted to reach standard form.

Common Student Mistakes

-  Leaving more than one non-zero digit before the decimal point in the 'k' part (e.g., writing 45×10^3 instead of correctly simplifying to 4.5×10^4).
-  Using a positive exponent for small numbers less than 1, when it should be negative.

Memory Hook

'Big number = positive power of 10; Tiny number = negative power of 10.'

 Advanced Insights

Scientific calculators and computer systems internally often use a very similar 'floating-point' representation (a significant digit part plus a separate exponent part) to efficiently store and calculate with an enormous range of numbers, from the subatomic to the astronomical.

II Quick Recap — Topics Covered So Far

- Exponents & Laws of Exponents — An exponent shows how many times a base number is multiplied by itself, and a set of consistent laws governs how exponents combine during multiplication, division, and powers of powers.
- Expressing Large Numbers in Standard Form — Standard form (scientific notation) expresses very large or very small numbers compactly as a number between 1 and 10 multiplied by a power of 10.

Chapter 13 — End of Chapter Revision

All Memory Hooks

- Exponents & Laws of Exponents: 'Same base: Multiply → ADD exponents; Divide → SUBTRACT exponents; Power of power → MULTIPLY exponents.'
- Expressing Large Numbers in Standard Form: 'Big number = positive power of 10; Tiny number = negative power of 10.'

All Common Mistakes

-  Adding exponents when the bases are actually different (the laws only apply when the base is the SAME).
-  Forgetting that a negative exponent means 'take the reciprocal', not 'make the number negative'.
-  Leaving more than one non-zero digit before the decimal point in the 'k' part (e.g., writing 45×10^3 instead of correctly simplifying to 4.5×10^4).
-  Using a positive exponent for small numbers less than 1, when it should be negative.

Formula Sheet

Formula	Meaning / Use
Laws of Exponents (Same Base): $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$	Whenever combining, dividing, or raising powers that share the same base.

Quick Revision Section

- Exponents & Laws of Exponents: An exponent shows how many times a base number is multiplied by itself, and a set of consistent laws governs how exponents combine during multiplication, division, and powers of powers.
- Expressing Large Numbers in Standard Form: Standard form (scientific notation) expresses very large or very small numbers compactly as a number between 1 and 10 multiplied by a power of 10.

Chapter 14: Symmetry

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic idea of line symmetry from Class 6.
- Ability to visualise rotating a shape around a fixed point.

14.1 Lines of Symmetry for Regular Polygons

 **Estimated time:** 12 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

A regular polygon (all sides and angles equal) has a number of lines of symmetry exactly equal to its number of sides, extending the line symmetry concept from Class 6 more formally.

Real-Life Example

A stop sign (regular octagon) has 8 lines of symmetry, while a regular hexagon (like a honeycomb cell) has exactly 6.

Detailed Explanation

Definition

A regular polygon is a closed figure with all sides and all angles equal; its lines of symmetry are the lines along which the figure can be folded so both halves match exactly.

Key Idea

For any regular polygon with n sides, it has exactly n lines of symmetry — this consistent pattern makes it easy to predict the number of symmetry lines just from the shape's name.

Working Principle

For regular polygons with an even number of sides, symmetry lines pass either through opposite vertices or through midpoints of opposite sides; for those with an odd number of sides, each symmetry line passes through one vertex and the midpoint of the opposite side.

Applications

- Design and architecture, where regular polygon symmetry is used for aesthetically pleasing, balanced patterns.
- Understanding tessellations and repeating geometric patterns in art and nature.

✓ Points to Remember (Exam Focus)

- ✓ Equilateral triangle (3 sides): 3 lines of symmetry.

- ✓ Square (4 sides): 4 lines of symmetry.
- ✓ Regular pentagon (5 sides): 5 lines of symmetry.
- ✓ Regular hexagon (6 sides): 6 lines of symmetry.
- ✓ General rule: a regular polygon with n sides has exactly n lines of symmetry.

⚠ Common Student Mistakes

- ⚠ Assuming an irregular polygon (with unequal sides/angles) also has this same number of symmetry lines — this rule only applies to REGULAR polygons.
- ⚠ Missing some lines of symmetry, especially in higher-sided regular polygons where lines aren't just horizontal/vertical.

🧠 Memory Hook

'Regular polygon with n sides = exactly n lines of symmetry.'

🚀 Advanced Insights

As the number of sides in a regular polygon increases toward infinity, both the shape and its infinite lines of symmetry approach that of a perfect circle, which famously has infinite lines of symmetry (every diameter is one).

14.2 Rotational Symmetry

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🗣 Quick Summary

A figure has rotational symmetry if it looks exactly the same after being rotated by some angle less than a full 360° turn around a fixed central point.

🌐 Real-Life Example

A ceiling fan with three identical blades looks exactly the same after being rotated by 120° around its centre, showing rotational symmetry.

📖 Detailed Explanation

Definition

Rotational symmetry exists when a figure, after being rotated by a certain angle (less than 360°) about a fixed point (the centre of rotation), appears identical to its original position; the order of rotational symmetry is the number of times the figure matches itself during one complete 360° rotation.

Key Idea

The order of rotational symmetry and the angle of rotation are directly related: $\text{angle of rotation} = 360^\circ / \text{order}$ — knowing one lets you immediately calculate the other.

Working Principle

To find the order of rotational symmetry, systematically rotate the figure by small increments around its centre and count how many distinct positions (including the starting position) make it look identical to the original.

Applications

- Design of wheels, fans, flags, and logos that rely on rotational symmetry for balance and visual appeal.
- Understanding natural patterns, like the petal arrangement in many flowers, which often show rotational symmetry.

✓ Points to Remember (Exam Focus)

- ✓ Order of rotational symmetry = $360^\circ \div \text{angle of rotation}$.
- ✓ A figure with order 1 has no real rotational symmetry (it only matches itself after a full 360° turn, same as any shape).
- ✓ A regular polygon with n sides has rotational symmetry of order n , with each rotation angle = $360^\circ/n$.
- ✓ Some figures have BOTH line symmetry and rotational symmetry (like a square), while others may have only one type.

⚠ Common Student Mistakes

- ⚠ Confusing the order of rotational symmetry (how many matching positions) with the angle of rotation (the specific angle for each match) — they are related but different values.
- ⚠ Assuming every figure with line symmetry must also have rotational symmetry, or vice versa — the two types are independent properties.

🧠 Memory Hook

'Order = $360^\circ \div \text{angle}$ — the two are always linked by this simple division.'

🚀 Advanced Insights

Some figures, like the letter 'S' or a pinwheel shape, have rotational symmetry but NO line symmetry at all — this shows the two types of symmetry are genuinely independent concepts, not simply two versions of the same idea.

Formula Section**📐 Order of Rotational Symmetry****Difficulty**

● Medium ★★☆☆

Formula	Order = 360° / Angle of rotation
Meaning of Variables	Order = number of times the figure matches itself in one full rotation, Angle of rotation = smallest angle needed for the figure to look identical again.
When to Use	To calculate either the order or the rotation angle of a figure's rotational symmetry, given the other value.
Common Mistakes	Forgetting that the angle used must be the SMALLEST angle that restores the original appearance, not just any matching angle.
Shortcut / Memory Trick	Find the smallest matching rotation angle first, then simply divide 360° by it to get the order.

II Quick Recap — Topics Covered So Far

- Lines of Symmetry for Regular Polygons — A regular polygon (all sides and angles equal) has a number of lines of symmetry exactly equal to its number of sides, extending the line symmetry concept from Class 6 more formally.
- Rotational Symmetry — A figure has rotational symmetry if it looks exactly the same after being rotated by some angle less than a full 360° turn around a fixed central point.

Chapter 14 — End of Chapter Revision

All Memory Hooks

- Lines of Symmetry for Regular Polygons: 'Regular polygon with n sides = exactly n lines of symmetry.'
- Rotational Symmetry: 'Order = $360^\circ \div \text{angle}$ — the two are always linked by this simple division.'

All Common Mistakes

-  Assuming an irregular polygon (with unequal sides/angles) also has this same number of symmetry lines — this rule only applies to REGULAR polygons.
-  Missing some lines of symmetry, especially in higher-sided regular polygons where lines aren't just horizontal/vertical.
-  Confusing the order of rotational symmetry (how many matching positions) with the angle of rotation (the specific angle for each match) — they are related but different values.
-  Assuming every figure with line symmetry must also have rotational symmetry, or vice versa — the two types are independent properties.

Formula Sheet

Formula	Meaning / Use
Order of Rotational Symmetry: Order = $360^\circ / \text{Angle of rotation}$	To calculate either the order or the rotation angle of a figure's rotational symmetry, given the other value.

Quick Revision Section

- Lines of Symmetry for Regular Polygons: A regular polygon (all sides and angles equal) has a number of lines of symmetry exactly equal to its number of sides, extending the line symmetry concept from Class 6 more formally.
- Rotational Symmetry: A figure has rotational symmetry if it looks exactly the same after being rotated by some angle less than a full 360° turn around a fixed central point.

Chapter 15: Visualising Solid Shapes

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic recognition of 3D shapes (cube, cuboid, cylinder, cone, sphere).
- Understanding of 2D polygons from earlier geometry chapters.

15.1 Faces, Edges, Vertices & Euler's Formula

Estimated time: 15 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

Every 3D solid (polyhedron) can be described by counting its flat faces, straight edges, and corner vertices, which are always related by a neat formula discovered by Euler.

Real-Life Example

A standard six-sided die (cube) has 6 faces, 12 edges, and 8 vertices — a simple, everyday object to verify Euler's formula.

Detailed Explanation

Definition

A face is a flat surface of a 3D solid; an edge is a line segment where two faces meet; a vertex is a corner point where edges meet; Euler's formula states that for any convex polyhedron, Faces + Vertices – Edges = 2.

Key Idea

Euler's formula ($F + V - E = 2$) holds true for ALL simple (convex) polyhedra, regardless of their specific shape or size, making it a powerful, universal check for consistency.

Working Principle

By carefully counting the number of flat faces, straight edges, and corner points of any convex 3D solid, you can verify Euler's formula, or use it to find a missing count if two of the three values are already known.

Applications

- Verifying the structural consistency of 3D models and designs in engineering and architecture.
- Basis for understanding more complex 3D geometry and topology studied in higher mathematics.

✓ Points to Remember (Exam Focus)

- ✓ Euler's formula: $F + V - E = 2$, valid for all simple (convex) polyhedra.
- ✓ Cube: 6 faces, 8 vertices, 12 edges ($6 + 8 - 12 = 2$, confirming the formula).
- ✓ Triangular pyramid (tetrahedron): 4 faces, 4 vertices, 6 edges ($4 + 4 - 6 = 2$).

✓ This formula does not directly apply to solids with curved surfaces (like a cylinder, cone, or sphere) in the same simple way.

⚠ Common Student Mistakes

- ⚠ Miscounting edges, especially in more complex solids, leading to an incorrect check of Euler's formula.
- ⚠ Trying to directly apply Euler's formula to curved solids like cylinders or cones without accounting for their curved surfaces properly.

🧠 Memory Hook

' $F + V - E = 2$ — Euler's magic constant for any simple solid.'

🚀 Advanced Insights

Euler's formula is one of the foundational results in a branch of mathematics called topology, which studies properties of shapes that remain unchanged even when the shape is stretched or bent (but not torn) — it's far more than just a geometry classroom fact.

Formula Section

📐 Euler's Formula for Polyhedra

Difficulty	● Medium ★★☆☆
Formula	$F + V - E = 2$
Meaning of Variables	F = number of faces, V = number of vertices, E = number of edges of a convex polyhedron.
When to Use	To verify the consistency of a solid's face/vertex/edge count, or to find one missing value given the other two.
Common Mistakes	Miscounting the edges or vertices of a complex solid, which throws off the entire formula check.
Shortcut / Memory Trick	Count faces and vertices first (usually easier), then use the formula to instantly verify or find the number of edges.

15.2 Nets, Views & Cross-Sections of Solids

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🗨 Quick Summary

A net is a flattened 2D pattern that folds up to form a 3D solid; solids can also be represented using front/side/top views, or by examining their cross-sections when sliced.



Real-Life Example

A cardboard cereal box, before being folded and glued, is essentially a flattened net of a cuboid; an architect's building plan often shows front, side, and top views of the same structure.



Detailed Explanation

Definition

A net is a two-dimensional flattened outline of a 3D solid, which can be folded along specific edges to construct that solid; a cross-section is the 2D shape obtained when a solid is sliced through by a flat plane.

Key Idea

The same 3D solid can often be represented by MULTIPLE different valid nets, and different flat views (front, side, top) capture different, complementary information about a single 3D object.

Working Principle

To identify if a given flat pattern is a valid net for a specific solid, mentally (or physically) fold it along the marked edges and check if all faces meet correctly to form the closed, complete solid shape.

Applications

- Packaging design, where efficient, correctly-folding nets are essential for manufacturing boxes and containers.
- Technical/engineering drawing, which relies on front, side, and top views to fully specify a 3D object on paper.

✓ Points to Remember (Exam Focus)

- ✓ A cube typically has 11 distinct possible nets (different valid flattened arrangements of its 6 square faces).
- ✓ Cross-sections of a solid depend on the angle and position of the cutting plane (e.g., a cylinder cut horizontally gives a circle; cut vertically through its axis gives a rectangle).
- ✓ Front, side, and top views together give a more complete picture of a 3D object than any single view alone.
- ✓ Not every arrangement of a solid's faces forms a valid, correctly-folding net.

⚠ Common Student Mistakes

- ⚠ Assuming there is only ONE correct net for a given solid, when in fact many solids (like a cube) have several valid net arrangements.
- ⚠ Ignoring the specific angle of a cutting plane when determining a cross-section's shape, leading to an incorrect answer.

**Memory Hook**

'Net = flatten to fold; Cross-section = slice to see inside; Views = look from different sides.'

**Advanced Insights**

This skill of translating between 2D representations (nets, views, cross-sections) and 3D solids is fundamental to fields like industrial design, architecture, and engineering, where 3D objects must always ultimately be communicated and manufactured using 2D drawings and plans.

II Quick Recap — Topics Covered So Far

- **Faces, Edges, Vertices & Euler's Formula** — Every 3D solid (polyhedron) can be described by counting its flat faces, straight edges, and corner vertices, which are always related by a neat formula discovered by Euler.
- **Nets, Views & Cross-Sections of Solids** — A net is a flattened 2D pattern that folds up to form a 3D solid; solids can also be represented using front/side/top views, or by examining their cross-sections when sliced.

Chapter 15 — End of Chapter Revision

All Memory Hooks

- Faces, Edges, Vertices & Euler's Formula: 'F + V - E = 2 — Euler's magic constant for any simple solid.'
- Nets, Views & Cross-Sections of Solids: 'Net = flatten to fold; Cross-section = slice to see inside; Views = look from different sides.'

All Common Mistakes

-  Miscounting edges, especially in more complex solids, leading to an incorrect check of Euler's formula.
-  Trying to directly apply Euler's formula to curved solids like cylinders or cones without accounting for their curved surfaces properly.
-  Assuming there is only ONE correct net for a given solid, when in fact many solids (like a cube) have several valid net arrangements.
-  Ignoring the specific angle of a cutting plane when determining a cross-section's shape, leading to an incorrect answer.

Formula Sheet

Formula	Meaning / Use
Euler's Formula for Polyhedra: $F + V - E = 2$	To verify the consistency of a solid's face/vertex/edge count, or to find one missing value given the other two.

Quick Revision Section

- Faces, Edges, Vertices & Euler's Formula: Every 3D solid (polyhedron) can be described by counting its flat faces, straight edges, and corner vertices, which are always related by a neat formula discovered by Euler.
- Nets, Views & Cross-Sections of Solids: A net is a flattened 2D pattern that folds up to form a 3D solid; solids can also be represented using front/side/top views, or by examining their cross-sections when sliced.